

Talent Scheduling with Daily Capacity and Changeover Times

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Outline

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What is Talent Scheduling?

Suppose we are filming a movie.

- Movies have scenes
- Each scene requires certain actors
- Each actor has to get paid for each day they're on set (even if they're not needed)

Goal: Minimize holding cost of actors by arranging the order of scenes filmed

Originally formulated in Optimal Scheduling in Film Production to Minimize Talent Hold Cost by T.C.E. Cheng, J.E. Diamond, and B.M.T Lin¹ in 1993 (1).

¹<https://doi.org/10.1007/BF00940554>

Talent Scheduling

Suppose we have a set of $n \in \mathbb{Z}^+$ *scenes* and a set of $m \in \mathbb{Z}^+$ *actors* and each scene requires **specific** actors. Additionally, each actor has a daily holding cost associated with them and each scene takes one day to film.

How do we film the set of scenes in a way that minimizes the holding cost of actors?

Consider: If Scene 1 is filmed on Day 1 and Scene 2 is filmed on Day 3, and Actor 1 is required for both scenes, Actor 1 needs to be paid for all 3 days.

But if we film Scene 1 on Day 1 and Scene 2 on Day 2, we will only need to pay Actor 1 for 2 days.

Example



Figure:
Bottoms(2023),
Dir. Emma
Seligman

Actors: {Rachel Sennott (PJ), Ayo Edebiri (Josie), Ruby Cruz (Hazel), Havana Rose Liu (Isabel), Nicholas Galitzine (Jeff)} = {Actor 1, Actor 2, Actor 3, Actor 4, Actor 5}

Scenes: {PJ and Josie hit Jeff with a car, Hazel talks to Josie and PJ about starting a "fight club", Josie and Isabel getting together, The "Fight Club" saves the football team} = {Scene 1, Scene 2, Scene 3, Scene 4}

Holding Costs: $\mathbf{c} = [1000, 1000, 800, 700, 500]$ (in USD)

Construct a binary
 $m \times n$ matrix to represent which actors are in which scenes:

$$T = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Example (cont.)

$$\mathbf{c} = [1000, 1000, 800, 700, 500] \quad T = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Optimal filming schedule:

Scene 1 (\$2000) $\rightarrow$ Scene 3 (\$1700) $\rightarrow$ Scene 2 (\$2800) + Holding (\$700) $\rightarrow$ Scene 4 (\$4000) = \$11,200

If we swap Scenes 2 and 3:

Scene 1 (\$2000) $\rightarrow$ Scene 2 (\$2800) $\rightarrow$ Scene 3 (\$1700) + Holding (\$800) $\rightarrow$ Scene 4 (\$4000) = \$11,300

Why Consider Daily Capacity and Changeover Times?

Assumptions:

- One day can only fit one scene
- Different scenes require different makeup / sets

Consider the following:

- Multiple scenes can be filmed in one day
- Makeup/set adjustments between each scene
- We cannot (legally) film all day

Previously, these things were not considered.

This was formulated by Zhaoqi Yang, Shunji Tanaka, Bertrand M.T. Lin² in Talent Scheduling with Daily Working Capacity and Scene Changeover Times in 2026 (2).

²<https://doi.org/10.1016/j.omega.2025.103439>

Daily Capacity and Feasible Sequences of Scenes

We can only film for so long each day. Therefore we can only take into account sequences of scenes whose durations and changeover times fit in a working day capacity, which we denote as W .

The duration of each scene j is stored in a vector $\mathbf{l} = (\ell_j) \in \mathbb{R}^n$, while the changeover time from scene i to j is prescribed by the asymmetric matrix $S = (s_{ij}) \in \mathbb{R}^{n \times n}$.

Thus, a possible approach is to consider all the possible sequences of scenes which satisfy the daily capacity constraint.

Example: If $\ell_1 + s_{12} + \ell_2 + s_{23} + \ell_3 \leq W$, then {Scene 1, Scene 2, Scene 3} is an admissible subset.

Finding Admissible Patterns

Define:

H - maximum number of scenes in a day

Our Method to Find Admissible P:

- (1) Obtain a list of permutations of n elements of max length H
- (2) Eliminate permutations that take too long (*total duration* $> W$)
- (3) Turn list of permutations into a set of sets

This returns a set of subsets of permutations that can fit in one day.

```
from itertools import permutations
def finding_P(n, S, durations, W, H):
    valid_perms = []

    for length in range(H):
        for perm in permutations(range(n), length+1):
            total = 0

            for scene in perm:
                total += durations[scene]

            for i in range(len(perm)-1):
                total += S[perm[i], perm[i+1]]

            if total <= W:
                valid_perms.append(perm)

    return valid_perms
```

Figure: Code to find P

Mathematical Formulation

This problem of talent scheduling can be framed in the context of Integer Linear Programming (ILP).

Objective: minimize $\sum_i c_i \cdot (\text{holding days of actor } i)$

Constraints:

- Every scene appears in *exactly one* pattern on *exactly one* day
- No gaps between filming days
- Actors are on set for all days needed until their required scenes are covered

Example



Figure: The Dark Knight (2008),
Dir. Christopher Nolan

Scenes: {Batman interrogates Joker, Joker blows up a hospital, The opening bank robbery, Joker at the Wayne Fundraiser}

Actors: {Christian Bale, Heath Ledger, Aaron Eckhart, Crew}

Scene Durations = [50, 30, 60, 80]

Holding Costs (in \$1000): $c = [1000, 1200, 800, 1300]$

Daily Work Capacity: $W = 200$

$$S = \begin{bmatrix} 0 & 100 & 10 & 10 \\ 40 & 0 & 60 & 50 \\ 40 & 90 & 0 & 30 \\ 20 & 60 & 20 & 0 \end{bmatrix} \quad \text{Availability: } T = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

Example

We first find our admissible P set.

We find that we can film a maximum of 3 scenes in a day. So, P will have elements with orders 1, 2, and 3. Let's find an element of P .

Scene1:
(50, {Actors})

Scene2:
(30, {Actor2,
Actor3, Crew})

Scene3:
(60, {Actor2, Crew})

Scene4:
(80, {Actors})

Example

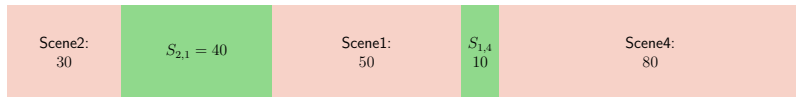
Suppose we first film Joker blowing up a hospital (Scene 2).



Next, we film the interrogation scene (Scene 1):



Let's try adding the longest scene, Scene 4:

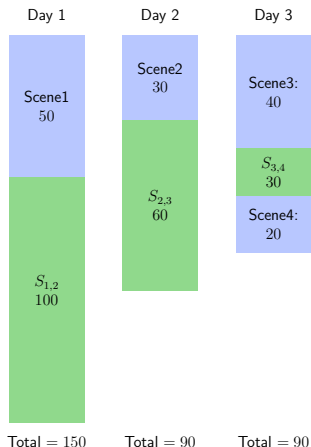


Example

We find the set of admissible P to be:

$\{\{3\}, \{3, 4\}, \{2, 3\}, \{1, 4\}, \{1, 2, 3\}, \{1, 2\}, \{1, 3\}, \{4\}, \{2\}, \{2, 4\}, \{1\}\}$

Putting all the information into the model, we get the solution:



- $\text{durations} = [50, 30, 60, 80]$

- $\mathbf{c} = [1000, 1200, 800, 1300]$

- $W = 200$

- $S = \begin{bmatrix} 0 & 100 & 10 & 10 \\ 40 & 0 & 60 & 50 \\ 40 & 90 & 0 & 30 \\ 20 & 60 & 20 & 0 \end{bmatrix}$

- $T = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$

Further Considerations

How can this problem/model be applied in other fields?

- Computer processing and task assignment
- Itinerary planning
- Patient/Doctor scheduling

How can we further modify this problem to simulate more realistic scenarios?

- Allowing for one scene to be shot on multiple days
- Forced holidays after mapping filming schedule to calendar
- What if accidents happen on /off set?
- What if instead of day rates we had hourly pay?

References

- [1] T. CHENG, J. DIAMOND, AND B. LIN, *Optimal scheduling in film production to minimize talent hold cost*, Journal of Optimization Theory and Applications, 79 (1993), pp. 479–492.
- [2] Z. YANG, S. TANAKA, AND B. M. LIN, *Talent scheduling with daily working capacity and scene changeover times*, Omega, 138 (2026).